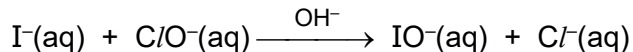


- 11 The kinetics of the reaction below were studied using the initial rate method.



experiment	initial $[\text{I}^{-}]$ / mol dm^{-3}	initial $[\text{C/O}^{-}]$ / mol dm^{-3}	initial $[\text{OH}^{-}]$ / mol dm^{-3}	initial rate / $\text{mol dm}^{-3} \text{ s}^{-1}$
1	0.0013	0.012	0.100	9.4×10^{-3}
2	0.0026	0.012	0.100	18.7×10^{-3}
3	0.0013	0.018	0.100	14.0×10^{-3}
4	0.0026	0.018	0.050	c

The unit of k is found to be s^{-1} .

Which statements are correct?

- 1 The order of reaction with respect to $[\text{OH}^{-}]$ is -1 .
 - 2 OH^{-} functions as a catalyst in this reaction.
 - 3 The value of c is 56.0×10^{-3} .
- A** 1, 2 and 3
- B** 1 and 3 only
- C** 2 and 3 only
- D** 3 only